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Final Report

The Art of Scientific Computing: Project Title: Respiratory Rates

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Chapter 1

Introduction

1.1 Overview

The respiratory rate is an important measure of a person's health and diagnosis of illnesses, defined as the number of completed breath cycles taken per minute. [1][2] While respiratory rates can be affected by non-disease related factors such as exercise, emotion, elevation and demographics [3][4], in this study we are interested in Apnoea. Apnoea (commonly known as sleep apnea) is an increasingly common disorder characterised by repeated airway collapses during sleep, which can lead to oxygen desaturation and disrupted sleep. [5] However, the measurement of respiratory rates directly can be an invasive , expensive and at times an inaccurate process. This has led to the underdiagnosis of Apnoea, as the diagnosis of Apnoea requires respiratory measurements.
This has motivated research into deriving respiration from Electrocardiograms (ECGs), as these measurements are low-cost and non-invasive [2]. The many methods which have been investigated over the years are broadly known as elecrocardiogram-derived respiration (EDR).

In this study, we first determine how to visualise respiratory data stored in WFDB format. Then, we investigate a method of deriving respitatory rates through ECGs via 4th order central moments. We then compare the derived respiratory rate to known datasets using spectrograms and regression analysis.

The most promising results came when comparing the spectrograms, where similarities in the patterns of the spectrogram were observed. Regression analysis showed little statistical correlation in the derived average respiration rates via EDR and experimental observations of chest respiratory efforts, showing that within the limited scope of this study, respirations rates derived via EDR have little predictive power on true respiration rates. If the predictive power can be improved, then it may be possible to incorporate machine learning to develop a possible pipeline to diganose regions of apnoea in an EDR-based pipeline.

1.2 Electrocardiograms

Electrocardiograms (ECGs) are measurements of your heart's electrical activity over time. Each beating of the heart produces a distinct pattern consisting of the P-wave, the QRS complex and the T-wave. The most relevant is the QRS complex, distinguished by the tall R-peak.

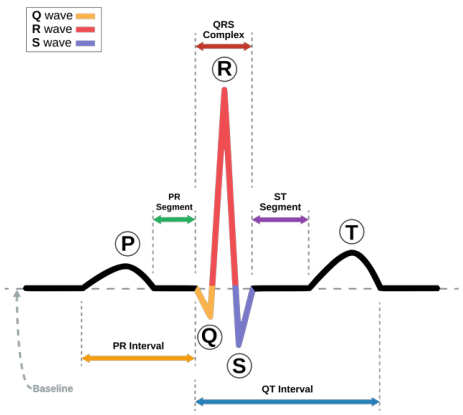


Figure 1.1: Sinus rhythm pattern showing the main components of an ECG waveform, including the QRS complex. Diagram courtesy of Joel Valerian.

A healthy person's heart rate increases and decreases slightly when breathing in and out, respectively, in a process known as Respiratory Sinus Arrhythmia (RSA) [6]. Despite the unnerving 'arrhythmia' in its name, RSA is in fact a sign of good health. These natural variations are the basis in which we can obtain information about respiratory efforts based on ECGs [7]. If accurate information about breathing rates can be extracted, it offers a low-cost and cheap alternative to direct respiratory measurements, a process which is significantly more invasive and expensive.

1.3 Reading WFDB files

ECG readings are typically stored in the WFDB format, where a single record consists of multiple files:

- .dat files contains the waveform data;
- .hea contains the header and information required to convert the waveform into the readings;
- .apn contains annotations for apnoea; and
- .qrs contains annotations for detected QRS complexes.

For example, the header file for record `a01` reads:

```
a01 1 100 2957000
a01.dat 16 200 12 0 -12 5827 0 ECG
```

The important information are:

- The sampling frequency f_s is 100Hz
- An ADC gain of 200mV/unit
- 295700 samples
- 12-bit ADC resolution. This means the amplitude is split into 4096 levels.

Therefore, the conversions for `a01` are:

$$\text{Amplitude (mV)} = \frac{\text{.dat Reading}}{200}$$

and

$$\text{Time (s)} = \frac{\text{.dat index}}{100}$$

In fact, most of the measurements in the database have the same ADC gain and sampling frequency, and so will have the same formula for conversion.
The typical .apn file will have an annotation of either "A" (for apnoea) or "N" (for normal), and the measurements are taken once every minute. .qrs files contain indices 'time-stamps' for when the QRS complex is detected, which are converted into time via the above formula.

1.4 Methodology

The data used in this study is from Pollard et al. [8] [9]. The database contains 70 ECG signals, which consist of a learning set of 35 records and a test set of 35 records. The distinction is that the learning set contain the .apn annotation file.
The typical pipeline for deriving respiratory signals from ECGs is shown in figure 1.2 [10]. The pre-processing is there to ensure that high-frequency noise is removed from the ECG, after which an ECG-derived respiratory (EDR) method using fourth-order central moments proposed by Schmidt et al. [11] is used. The fourth-order central moment is defined as

$$m_4 = \frac{1}{L}\sum_{i=1}^{L}(x_i - \mathbb{E}[X])^4$$

where $X = [x_1, x_2, \ldots, x_L]$ is a segment of the data of length $L = 0.02 \times f_s$ with sampling frequency f_s. In this method, one takes the fourth-order central moment of the ECG and interpolates a cubic spline over the local maxima. The result, after post-processing, is an EDR signal, which is saved in a separate file for convenient access later.

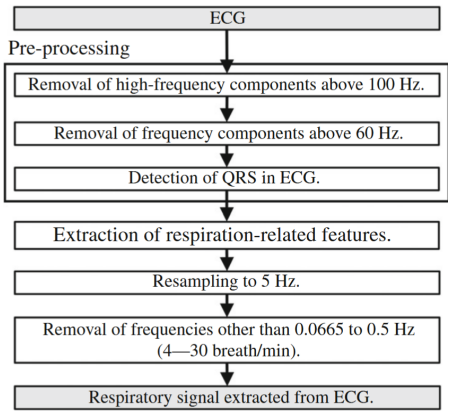


Figure 1.2: The typical pipeline for deriving respiratory signals from ECGs. Note that Akai et al. used signals with a frequency of 1000Hz, so exact values for pre-processing may differ from study to study. Diagram from Akai et al. [10].

The language used is Python, and the code for calculating the EDR is `EDR_Signal_Generator.ipynb`. The numpy arrays for both the EDR and time were saved in .npz format - a zipped archive format used to store numpy arrays. The code was able to convert and save the EDR and time at an average of about 6 seconds per record. The time taken is heavily dependent on the length of the record, which can range from 7 to 10 hours [8].

The obtained EDR is itself not a direct measure of the respiratory rate, but rather the respiratory effort. The breathing rate is estimated by performing a Short Time Fourier Transform (STFT) with a time window of 30 seconds. Then, the breathing rate over a 30-second period is the frequency with the highest peak in the spectrogram in that time period.

The database also provides additional respiratory measurements for select records, which include Chest and Abdominal respiratory effort signals obtained using respiratory inductance plethysmography (RIP), a technique which estimates changes in lung volume by non-invasively measuring contractions and expansions of the adbomen and thorax [12]. We use these measurements as validation for the EDR derivation method that we described above.

Plotting the spectrogram of the EDR can reveal visual patterns which could provide 'tells' for Apnoea. We will also use spectrograms to compare the EDR with experimental measurements of respiration efforts.

1.5 Computer code

Download `Project_Software.zip` for the code and data. In order to run the code, you must have the following packages installed in your Python library:
`wfdb, numpy, matplotlib, pathlib, math, scipy, time, pandas, statsmodels`

Chapter 2

Findings

2.1 Results and Discussion

Snapshots of electrocardiograms are as below:

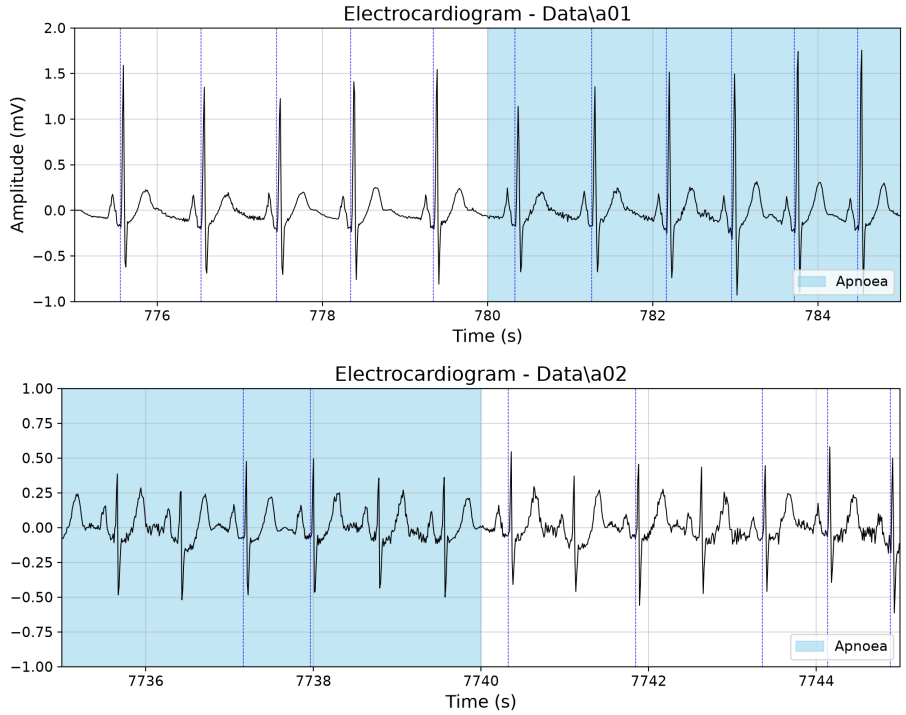


Figure 2.1: Snapshots of electrocardiograms for records `a01` and `a02`, respectively. The dotted blue lines are the QRS annotations provided by the database.

As is evident in the ECG for `a02` in 2.1, not all of the QRS complexes were annotated correctly by the provided annotation file. The difficulty may have arisen from the p-waves almost becoming as tall as the R-peak. The method used to find the peaks for the 4th order central moments was to find the 'mid-points' between QRS annotations and finding the maxima in-between those points. While this method is suffient when the QRS complexes are annotated correctly, if there are missing annotations, this causes peaks to be missed. The result is that there will appear to be regions where the respiratory rate will look slower than is actually the case.

A snapshot of the obtain EDR is shown in figure 2.2. As can be seen in (d), the frequency of the EDR appears to be higher than the measured respiratory efforts.

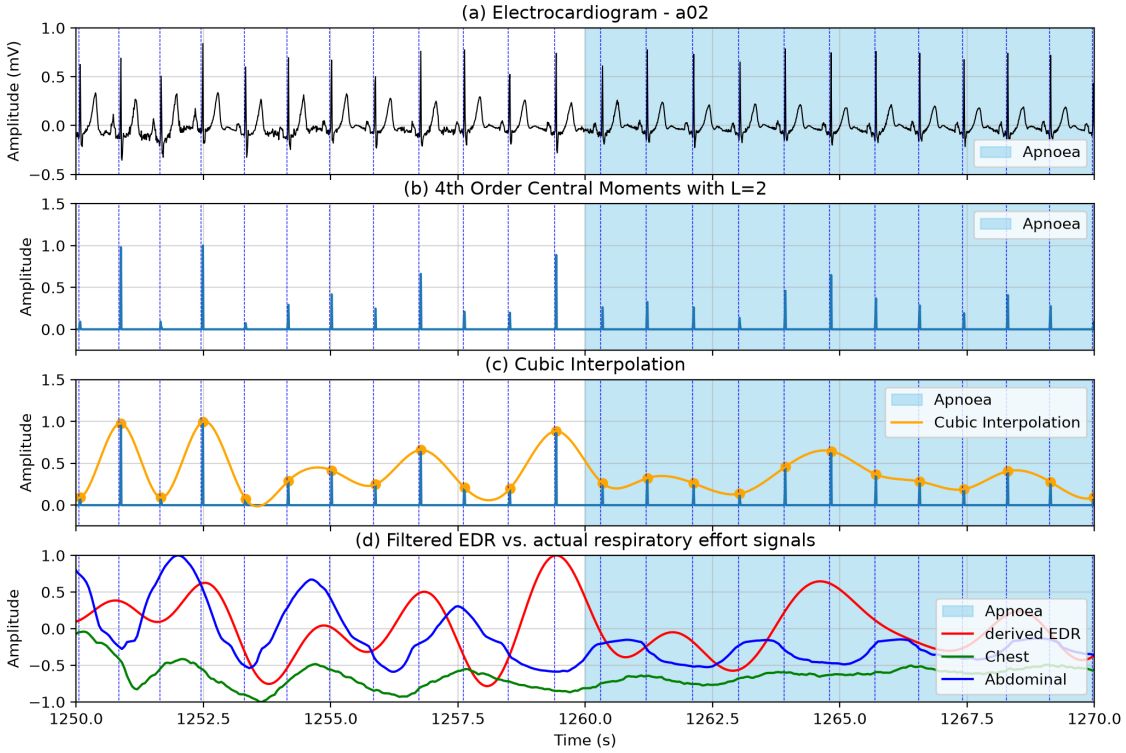


Figure 2.2: A visual guide to estimating the EDR signal, using record `a02` as an example. (a) Original ECG readings. (b) Calculating the 4th order central moments with L=2. (c) Performing a cubic interpolation on the peaks. (d) comparison between the EDG and measured respiratory efforts after post-processing.

2.1.1 Spectrogram

Note that all the findings and figures were generated from `STFT.ipynb`.
The spectrograms for EDRs and chest measurements can be seen in figure 2.3. For record `a01`, both spectrograms have clear 'bright bands' for frequencies between 15-25 beats/min, suggesting that the EDR was able to capture the overall respiratory rate well. On the other hand, for `a02`, the pattern on the spectrogram is more uniform. Furthermore, In the region $t \in [129, 176]$, most of the frequencies between 4-30 are absent, whereas in the chest measurements, there is a bright band at around 20 beats/min. Nonetheless, there seems to be evidence to suggest that when a person exits a state of apnoea and return to normal breathing, we sometimes observe decreases in intensities of many of the frequencies in the 4-30 breaths/min range. This could be due to the absence of QRS annotations, suggesting that the machine learning tool used by the database had difficulties in annotations during the transition from apnoea to normal breathing.

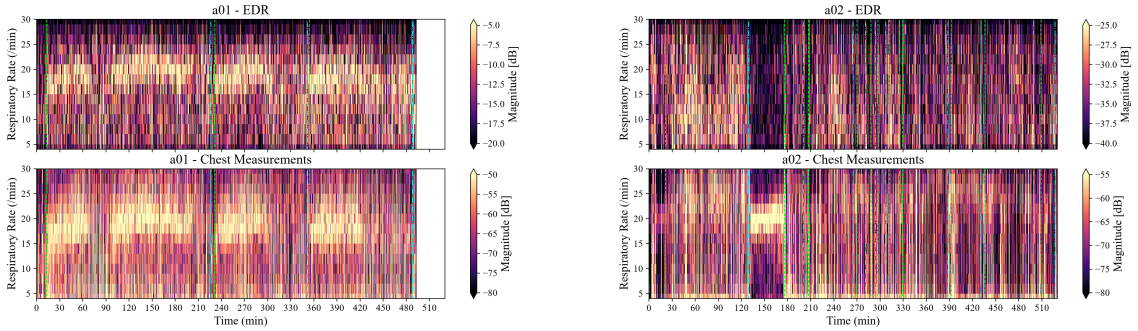


Figure 2.3: Spectrograms for EDR and chest measurements for `a01` and `a02`, respectively. The green line indicate onset of apnoea and light blue lines indicate a return to normal breathing.

When a line is traced through the brightest block in the spectrogram for each time period, one obtains an estimate of the average respiratory heart rate with a window of 30 seconds. In figure 2.4, we further averaged the respiratory rate with a rolling window of 2 minutes. In general, the more the respiratory rate is averaged out, the less variance the difference has.

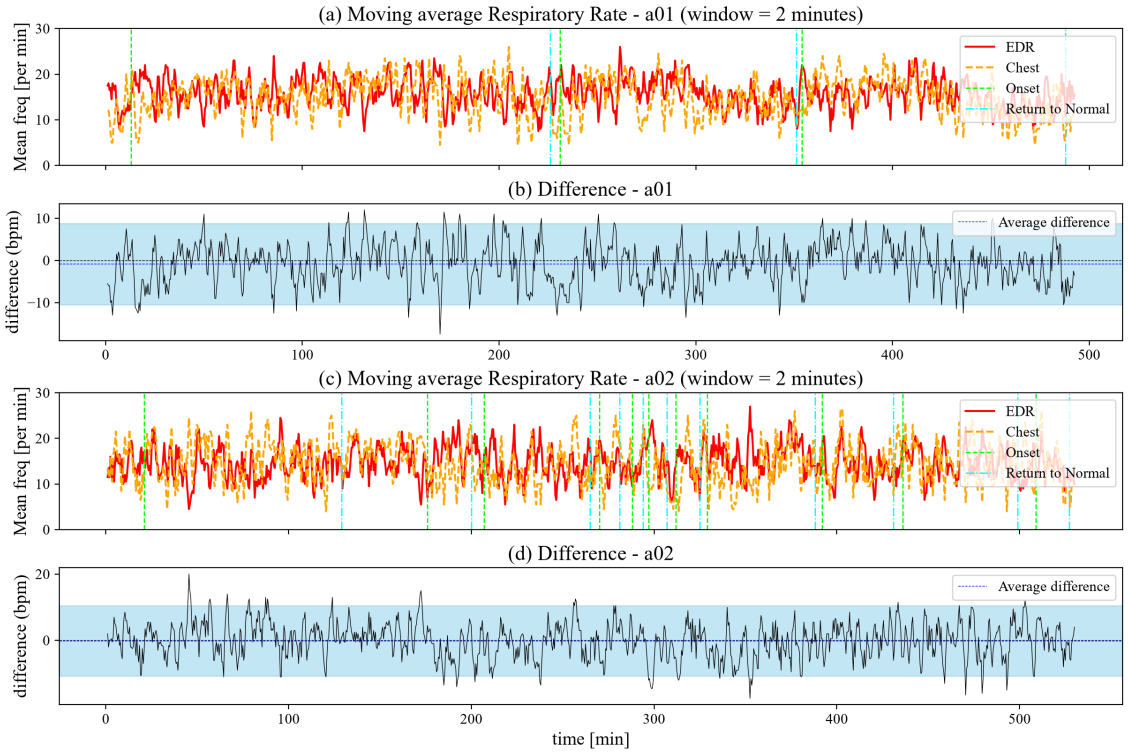


Figure 2.4: (a,c) Average respiratory rate, extracted from the respiratory efforts from the Chest and EDR, over a 2 minute window. (b,d) Difference between respiratory rate of chest minus EDR with the 95% confidence interval. `a01`: $\mu = -0.836890, \sigma = 4.894213$. `a02`: $\mu = -0.116147, \sigma = 5.428670$

2.1.2 Regression Analysis

To further understand how well the EDR predicts chest respiratory rate, we can perform a regression analysis. Let EDR_t and Chest_t denote the average respiratory rate derived from the EDR and RIP at time t, respectively. Then, we perform a regression analysis using `pandas` and `statsmodel` on the simple regression model

$$\text{Chest}_t \sim \beta_0 + \beta_1 \, \text{EDR}_t$$

with Chest_t being the dependent variable and EDR_t the explanatory variable. The result of the Ordinary Least Squares (OLS) regression is as below:

OLS Regression Results

Dep. Variable:	Chest	**R-squared:**	0.017
Model:	OLS	**Adj. R-squared:**	0.016
Method:	Least Squares	**F-statistic:**	32.96
Date:	Thu, 22 Oct 2026	**Prob (F-statistic):**	1.08e-08
Time:	23:14:14	**Log-Likelihood:**	-5876.2
No. Observations:	2043	**AIC:**	1.176e+04
Df Residuals:	2041	**BIC:**	1.177e+04
Df Model:	1		
Covariance Type:	HC1		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	12.2366	0.435	28.125	0.000	11.383	13.090
EDR	0.1620	0.028	5.741	0.000	0.107	0.217

Omnibus:	36.131	**Durbin-Watson:**	0.450
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	20.527
Skew:	-0.021	**Prob(JB):**	3.49e-05
Kurtosis:	2.511	**Cond. No.**	69.8

Note that the Standard Errors are heteroscedasticity robust (HC1), that is, the results of the regression analysis is accurate even when the standard errors are not assumed to be uniform for all observations.
The coefficients of interest are statistically significantly different from 0 at 5% confidence level. Based on this regression analysis, we can estimate respiratory rates from averaged EDR rates according to the formula

$$\text{Chest}_t = 12.2366 + 0.1620 \times \text{EDR}_t$$

Due to the scope of the research project, we were not able to do any further regression analysis, such as multiple linear regression or IV estimation. At this stage, we have not demonstrated that this model does not suffer from omitted variable bias, so the result should be interpreted causally.
Plotting the chest measurements against the EDR measurements reveals that there is low correlation (albeit statistically significant at 5% level) between the respiratory rate from EDR measurements and Chest measurements (See Figure 2.5). Indeed, the Pearson r value for the combined database of `a01` and `a02` for chest and EDR measurements came out to only a 0.1301, suggesting a weak statistical relationship.

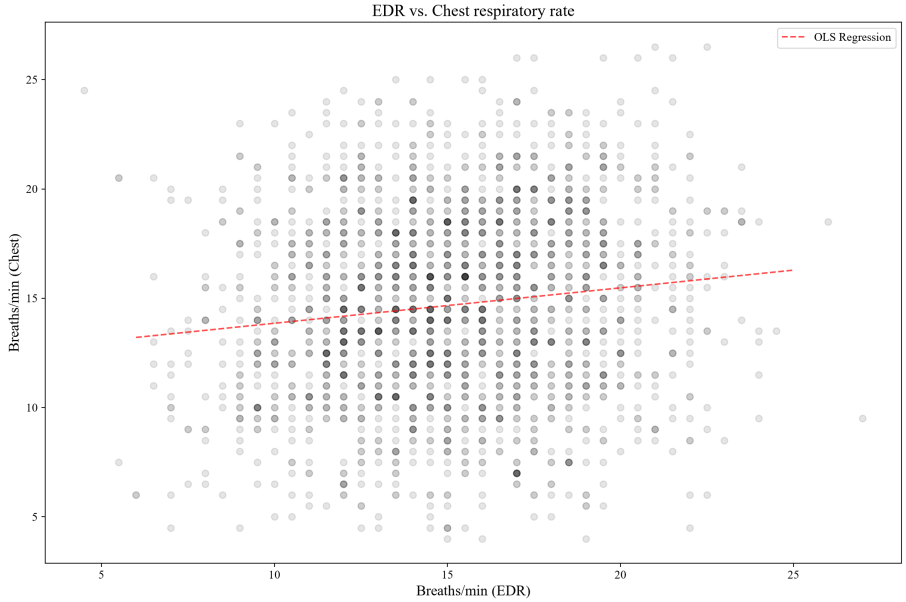


Figure 2.5: A scatter plot of respiratory rates from Chest vs. EDR. Opacity indicates frequency - the more frequent, the darker the dot. The red line is the simple regression model we obtained from 2.1.2.

The most significant bottleneck in the data which was used was the sampling frequency, as many similar studies have used data with much higher frequencies. This is problematic, as it is desirable for the window for 4th order central moments to be as wide as possible. In our study, we were limited to $L = 2$. Furthermore, The Butterworth low-pass filter is limited to a ceiling of 50Hz (Nyquist frequency), which meant I couldn't use that filter for the pre-processing. We ultimately filtered using a Fast Fourier Transform method.
Another significant challenge is that based on the database, it is not possible to distinguish the demographics of the person who was recorded (e.g. age, gender, other pre-existing health conditions), which are highly relevant in establishing what a 'typical' resting breath rate/heart rate is for such a person [3][4]. This, consequently, affects the respiratory rate.

2.2 Further directions

Other methods exist in the umbrella of EDR, which utilise values of e.g. Q-peak, R-peak, the gradient of the R-wave, and more. [10]. These methods are outside the scope of the study, but could be of interest in future studies.
The statistical analysis performed on the data was highly simplified, as we only looked at the correlation between true measurements and EDR measurements. It could have been worthwhile to inspect how the distribution of the respiratory rate differ for states under apnoea and with normal breathing differ, and whther they could offer insight into apnoea detection for un-annotated ECGs. It is also worth noting that since the respiratory rate is time-series data, running an ARDL(1,1) model such as

$$\text{EDR}_t = \beta_0 + \beta_1 \text{EDR}_{t-1} + \gamma_1 \text{Apn}_t + \gamma_2 \text{Apn}_{t-1} + \varepsilon_t$$

where Apn_t is 1 if apnoea is detected in the time-frame and 0 otherwise, could have unveiled information about how Apnoea affects the variability of the respiratory rate. Due to time constraints, both topics were beyond the scope of this study, but it is nonetheless a possible future direction for research.

2.3 Conclusion

In conclusion, we were able to develop softwares which can plot ECGs from WFDB files, convert records into respiratory data using EDR, and use STFT to plot spectrograms and approximate respiratory rates. Simple linear regression analysis showed little statistical correlation between respiraty rates derived from chest measurements and EDR methods. Further research could involve inspecting other EDR methods and comparing the accuracy, and more complex statistical analysis. Overall, it is clear that with more research, EDRs could offer a cheap yet accurate estimation of respiration patterns, which could increase the rate of diagnosis of respiration-related illnesses/condition, particularly in under-resourced regions, and save many lives.

Appendix A

Program flowchart

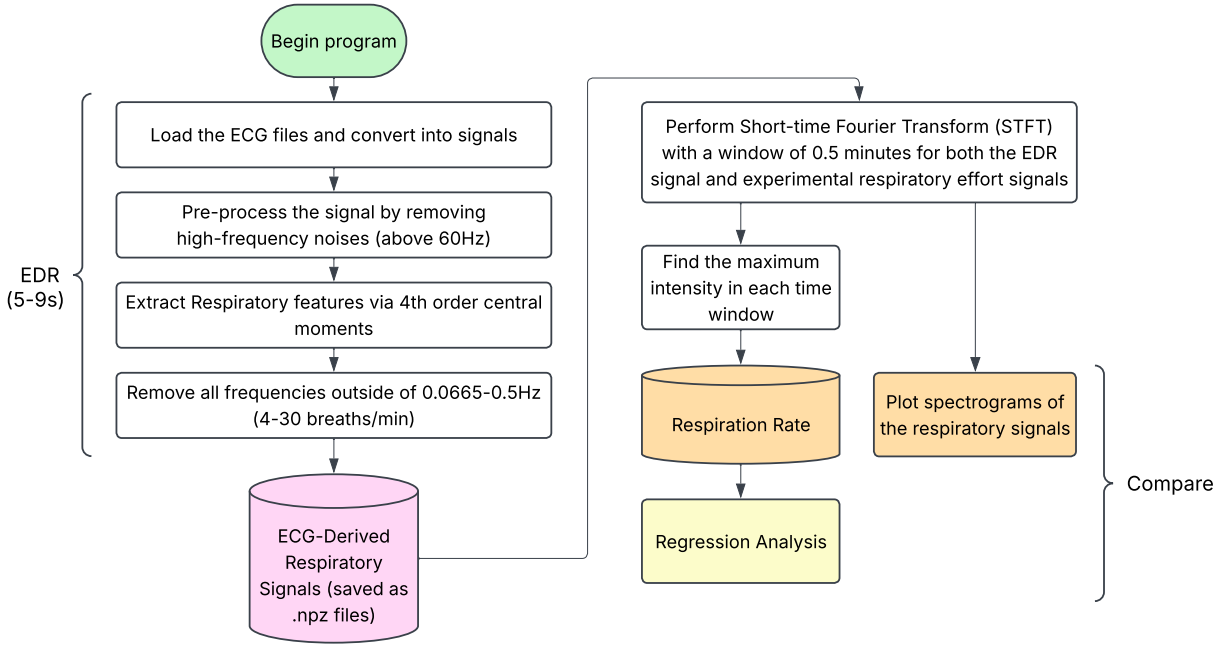


Figure A.1: A flowchart for the project, including the typical time taken (from our computers) to run the EDR. The number may be different on different software, and may take longer if the ECG data is lengthy.

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